Beyond Mean–Variance: Goal-Based Portfolio Optimisation with Derivatives and Structured Products

A reproducible computational study — replicating, correcting and scaling a mental-accounting optimiser

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Abstract

We document a reproducible optimiser that builds goal-based portfolios beyond the mean–variance frame. It re-implements a 2012 MSc thesis (USI Lugano) replicating the mental-accounting models of Das & Statman and Das, Markowitz, Scheid & Statman, and extends them in two directions. First, two corrections to the thesis: strict enforcement of the Expected-Shortfall constraint, and arbitrage-consistent (rather than par) pricing of structured notes, each reported with before/after figures. Second, a scalable Monte-Carlo + CVaR engine that escapes the grid's curse of dimensionality through copula scenarios and a linear program, pricing multiple derivatives consistently inside each scenario. We give a complexity result, a sample-average-approximation convergence theorem, and a proposition pinning down exactly when the two engines solve the same tail-constrained problem; measured scaling to 150 instruments; benchmarks against mean–variance, equal-weight and risk-parity; robustness to copula choice, sample size and estimation error; and a multi-regime out-of-sample backtest. The contribution is computational and methodological rather than a new financial theory; the novelty is the composition of a goals-based behavioural objective, a coherent tail-risk constraint, and per-scenario arbitrage-consistent derivative pricing at scale.

Keywords: behavioral portfolio theory, mental accounting, Das–Statman model, goal-based investing, Expected Shortfall, Conditional Value-at-Risk, derivatives, structured products, Monte-Carlo, copula, Rockafellar–Uryasev, out-of-sample backtesting.
JEL: G11, G41, G17, C61, C63.

1 Background and contribution

Markowitz mean–variance (MV) analysis treats risk as a single portfolio variance and assumes the investor optimises one aggregate trade-off. Behavioural evidence instead suggests that people manage wealth in mental accounts tied to distinct goals, each with its own attitude to loss. Das, Markowitz, Scheid and Statman (2010) formalise this and prove a precise equivalence between the mental-accounting (MAT) specification and classical MV; Das and Statman (2009, 2013) extend goal-based optimisation to derivatives and non-normal returns. The author's 2012 MSc thesis (USI Lugano, supervised by Prof. E. De Giorgi) replicated both methodologies in R.

This paper is explicit about its three-way contribution, and labels every result accordingly throughout (see the provenance table, Section 14):

- **Replication:** the thesis grid optimiser, re-implemented in Python.

*Companion to the interactive *Beyond Mean-Variance* optimiser. App: `https://sami-jeddou-behavioral-portfolio-optimizer.streamlit.app`; code: `https://github.com/SamiJeddou/behavioral-portfolio-optimizer`. Research and educational project; not investment advice.

- **Correction:** two documented fixes to the thesis behaviour — Expected-Shortfall enforcement and fair pricing of structured notes (Section 13).
- **Extension:** a scalable Monte-Carlo + CVaR linear-program engine with per-scenario arbitrage-consistent pricing of multiple derivatives (Sections 5–6), and live-data, multi-regime backtesting (Section 12).

We do not claim a new financial theory. The MVT–MAT equivalence is due to Das et al. (2010) and CVaR optimisation to Rockafellar and Uryasev (2000); the contribution is the *composition* of these into a goals-based, derivative-aware optimiser that scales to realistic mandates, with the theoretical and empirical support a reviewer expects.

2 The mental-accounting framework

An investor allocates wealth across goal sub-portfolios. Each goal carries a return threshold H and a maximum acceptable probability α of falling short of it. With weights w on the simplex $\Delta = \{w : w^\top \mathbf{1} = 1,\, w \geq 0\}$ and portfolio return $r_p = w^\top r$, the Value-at-Risk form of the goal problem is

$$\max_{w \in \Delta} w^\top \mu \quad \text{s.t.} \quad \mathbb{P}(r_p \leq H) \leq \alpha. \tag{1}$$

A tail-average (Expected-Shortfall) variant replaces the shortfall probability with a floor on the conditional mean of the loss tail, $\mathbb{E}[r_p \mid r_p < H] \geq L$. The key theoretical result is the MVT–MAT equivalence: each goal's optimum is mean–variance efficient and can be written as $\max_w w^\top \mu - \frac{\lambda}{2} w^\top \Sigma w$ for an implied risk-aversion λ fixed by the binding shortfall constraint. The optimiser reports an implied λ for every goal to reconcile the two views.

3 The exact grid optimiser

Asset returns are modelled as multivariate normal with mean μ and covariance Σ. The engine enumerates candidate portfolios on a discretised grid: each asset's return axis is discretised into m points spanning $\mu \pm k\sigma$, and portfolio weights are searched on a simplex grid whose spacing corresponds to m' steps per weight axis. For n assets the state space grows as roughly m^n—the result is exact and thesis-faithful but practical only to about four assets; for five or more, a differential-evolution search replaces the exhaustive grid. Two Expected-Shortfall modes are offered: a *thesis-faithful* mode that reproduces the original R behaviour, and a *rigorous-ES* mode that maximises return strictly subject to the tail-average floor (Section 13).

4 Derivatives and structured products

Sixteen instruments are priced arbitrage-consistently with Black–Scholes—vanilla options, spreads, collars, capital-guaranteed and barrier notes, and certificates. The Black–Scholes call is

$$C(S_0, K, T) = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \tag{2}$$

Under each modelled outcome the instrument's payoff is evaluated and folded into the portfolio return, so asymmetric, non-normal payoffs enter the optimisation directly rather than through a variance proxy. A structured note is valued as the present value of its bond floor plus its embedded option(s), not at par (Section 13).

5 The scalable Monte-Carlo + CVaR engine

To escape the m^n state space, the second engine samples S joint return scenarios through a copula—Gaussian, or Student-t for tail co-movement—prices its supported derivatives under each scenario, and solves the goal as a linear program. Tail risk is measured coherently with Conditional Value-at-Risk. The Rockafellar–Uryasev representation is

$$\mathrm{CVaR}_\alpha(\ell) = \min_{\zeta\in\mathbb{R}} \Big\{\zeta + \tfrac{1}{\alpha}\,\mathbb{E}\big[(\ell-\zeta)_+\big]\Big\}, \tag{3}$$

which is linear in scenario form. Collect the per-scenario returns of all N securities and K derivatives into an $S\times(N{+}K)$ matrix R, each row an equiprobable joint scenario, and write the loss as $\ell_s = -w^\top R_s$. The mental-account problem $\max_w \mathbb{E}[r]$ s.t. $\mathrm{ES}_\alpha(r) \geq L$, $\mathbf{1}^\top w = 1$, $w \geq 0$ becomes the linear program

$$\begin{aligned} &\max_{w,\zeta,z}\ \widehat{\mu}^\top w \\ &\text{s.t. } \zeta + \frac{1}{\alpha S}\sum_{s=1}^{S} z_s \leq -L, \\ &\qquad z_s \geq -w^\top R_s - \zeta, \quad s=1,\dots,S, \\ &\qquad z_s \geq 0, \quad \mathbf{1}^\top w = 1, \quad 0\leq w\leq \bar{w}, \end{aligned} \tag{4}$$

with decision vector $x=(w,\zeta,z)\in\mathbb{R}^{N+K+1+S}$ and a sparse constraint matrix—each scenario row touches the $N{+}K$ weights, ζ, and one slack z_s. *Although a derivative's payoff is non-linear in its underlying, it is priced once per scenario and enters as a pre-computed constant column $R_{s,N+k}$; the portfolio return $w^\top R_s$ is therefore linear in the weights, which is precisely what keeps the problem a linear program.* Intuitively, ζ locates the Value-at-Risk level and each slack z_s measures how far the loss in scenario s exceeds that level, so the first constraint caps the average of the worst-α tail losses while the objective pushes expected return up. Position limits, group budgets and turnover penalties are additional linear rows. It is solved with the HiGHS LP solver, and a single set of common random numbers is reused across floors to give a smooth, comparable frontier; sweeping the floor L traces the return/tail-risk frontier with warm starts.

Scenario generation. With Cholesky factor L_c of the correlation matrix ($\Omega = L_c L_c^\top$), drawing $\xi_s \sim N(0, I_N)$ and setting $z_s = L_c\xi_s$, $R_{s,i} = \mu_i + \sigma_i z_{s,i}$ reproduces the multivariate-Normal assumption. A Student-t copula replaces the Gaussian dependence to capture assets falling together in stress, with marginals held fixed (Sklar's decomposition).

Per-scenario derivative pricing. Each derivative k on underlying $u(k)$ has entry price the Black–Scholes fair value π_k at $S_0 = 1$; in scenario s the terminal spot is $1 + R_{s,u(k)}$, the payoff $\Pi_k(1+R_{s,u(k)})$ follows from the instrument's leg structure, and the derivative's scenario return is

$$R_{s,N+k} = \frac{\Pi_k\big(1+R_{s,u(k)}\big) - \pi_k(1+p_k)}{\pi_k(1+p_k)},$$

with p_k an optional structured-product premium. Because the payoff is computed from the same simulated underlying that drives the security column, each scenario keeps a coherent $(\text{underlying}, \text{derivative})$ pair with no auxiliary model.

Multiple derivatives and mental accounts. K derivative columns are optimised jointly at no extra modelling cost, so the optimiser can fund downside protection on one holding while gearing upside on another. Each behavioural sub-portfolio (goal) is then the same linear program (4) over the same scenario matrix with its own triple (H_j, α_j, L_j); solving the accounts independently and aggregating preserves the goals-based interpretation while inheriting the scalability.

Instrument coverage. The full sixteen-instrument library—including the structured notes and the path-dependent barrier note—is available in the grid engine. The scalable engine currently covers single-underlying options with up to two strikes (vanilla calls/puts, straddles, strangles, vertical spreads); extending it to the static multi-leg payoffs is straightforward, the path-dependent ones less so. We treat this asymmetry as stated future work.

6 Theoretical properties

This section gives the formal support the engine's claims require.

Proposition 1 (Complexity separation)**.** *Let N securities be discretised at m levels in the grid engine, and let the scenario engine draw S scenarios over N securities and K derivatives. The grid enumerates $\Theta(m^N)$ joint states, exponential in N. The scenario program* (4) *has $N+K+1+S$ variables and $S+1$ constraints; assembly is $O(S(N+K))$ and each interior-point iteration is polynomial in the program size. The scenario reformulation is therefore linear in the number of instruments and independent of the discretisation m.*

Proof sketch. The grid is the Cartesian product of N axes of m points, hence m^N cells. The variable and constraint counts of (4) are read directly from the Rockafellar–Uryasev form; the constraint matrix has $O(S(N+K))$ nonzeros, and HiGHS' simplex/interior-point steps are polynomial in size. $\square$

Theorem 1 (Consistency of the scenario optimiser)**.** *Fix $\alpha \in (0,1)$ and floor L. Let returns R have distribution P with $\mathbb{E}\|R\| < \infty$ and a continuous loss c.d.f., and let $W = \{w : \mathbf{1}^\top w = 1,\ 0 \leq w \leq \bar{w}\}$ be compact. Let $v^\star$ be the optimal value of the population problem $\max_{w \in W} \mu^\top w$ s.t. $\mathrm{CVaR}_\alpha(-w^\top R; P) \leq -L$, and $\widehat{v}_S$, $\widehat{w}_S$ the optimum of the S-scenario sample problem under the empirical measure. If a Slater point exists (some $w \in W$ with strict feasibility), then almost surely as $S \to \infty$, $\widehat{v}_S \to v^\star$ and $\mathrm{dist}(\widehat{w}_S, W^\star(P)) \to 0$, where $W^\star(P)$ is the set of population optimisers. Moreover the empirical CVaR is asymptotically Normal, $\sqrt{S}\big(\mathrm{CVaR}_\alpha(\cdot; \widehat{P}_S) - \mathrm{CVaR}_\alpha(\cdot; P)\big) \Rightarrow N(0, \tau^2)$, so the optimal value converges at the $O(S^{-1/2})$ rate.*

Proof sketch. $\mathrm{CVaR}_\alpha(\cdot; P)$ from (3) is convex in w and jointly continuous in (w, P) under the first-moment condition. Over compact W the sample objective and constraint converge uniformly to their population counterparts by the uniform law of large numbers for the integrable Rockafellar–Uryasev integrand; Slater's condition supplies constraint qualification, so epi-convergence of the sample problems yields convergence of optimal values and (SAA consistency) of optimisers. Asymptotic Normality of the empirical CVaR is the Rockafellar–Uryasev (2002) estimator CLT. This is the standard sample-average-approximation result for CVaR-constrained convex programs (Shapiro, Dentcheva & Ruszczyński, 2009). $\square$

The result covers option overlays without modification: although a derivative payoff is non-smooth (kinked) in its underlying, it enters as a constant per-scenario column $R_{s,N+k}$, so the sample objective and CVaR constraint remain piecewise-linear and the convergence argument is unaffected. The consistency is to the optimum of the *CVaR* problem, not to the grid's *VaR/shortfall* optimum. The next result says exactly when the two coincide.

Proposition 2 (Grid-ES and CVaR coincidence)**.** *Let $r = w^\top R$ have a continuous, strictly increasing c.d.f. with α-quantile q_α. Then the grid's fixed-threshold tail mean and the α-CVaR coincide exactly iff the threshold is the α-quantile:*

$$\mathbb{E}[r \mid r < H] = \mathrm{ES}_\alpha(r) = -\mathrm{CVaR}_\alpha(-r) \quad \Longleftrightarrow \quad H = q_\alpha.$$

For $H \neq q_\alpha$ the two are valid but distinct tail measures, and since $\mathrm{CVaR}_\alpha \geq \mathrm{VaR}_\alpha$ *always, a CVaR floor is a conservative tightening of the corresponding shortfall constraint.*

Proof sketch. With a continuous c.d.f. and no atom at q_α, $\mathrm{ES}_\alpha(r) = \mathbb{E}[r \mid r \leq q_\alpha]$. Setting $H = q_\alpha$ equates the conditioning events $\{r < H\}$ and $\{r \leq q_\alpha\}$ up to a null set, so the conditional means agree; for $H \neq q_\alpha$ the events differ and $\mathbb{P}(r < H) \neq \alpha$. The inequality is standard. □

This proposition is the principled basis for keeping the grid (VaR) and scalable (CVaR) results in *separate* panels (Tables 7 and 8): they answer related but distinct questions, and are not directly comparable unless H is the α-quantile.

Value-at-Risk via a CVaR proxy. The thesis chance constraint $\mathbb{P}(r < H) \leq \alpha$ is itself non-convex and combinatorial (mixed-integer in general). Rather than solve it exactly, the engine uses CVaR as a conservative convex surrogate: since $\mathrm{CVaR}_\alpha \geq \mathrm{VaR}_\alpha$, a CVaR floor meets the VaR-style floor at least as safely, and ES is in addition the coherent measure favoured by modern regulation. Where an exact $\mathbb{P}(r < H) \leq \alpha$ solution is required, the convex CVaR optimum could serve as a warm start for a subsequent mixed-integer refinement, which this version does not implement.

7 Validation

Every figure is checked rather than asserted. Under the Gaussian copula the scenario Expected Shortfall matches the closed-form Normal value $\mathrm{ES}_\alpha = \mu_p - \sigma_p\, \phi(\Phi^{-1}(\alpha))/\alpha$: on a five-asset equal-weight portfolio the Monte-Carlo estimate is −18.70% versus the closed form −18.73% (3 basis points; $S = 4 \times 10^4$), with sample mean 0.0810 vs exact 0.0820 and σ_p 0.1301 vs 0.1305. The frontier is monotone, the ES floor binds at the optimum to machine tolerance, and infeasible floors are flagged rather than silently relaxed. The grid base case reproduces the thesis: at $H = -10\%$, $\alpha = 5\%$ the no-derivative behavioural optimum returns 9.78% with the Markowitz optimum coinciding with it (MVT–MAT equivalence), and the shortfall constraint binds exactly at 5.00%.

8 Scaling and complexity, measured

We confirm Proposition 1 empirically. Table 1 times the scenario LP as the number of instruments grows at fixed $S = 4{,}000$; Table 2 varies S at fixed $N = 20$. A 150-instrument behavioural portfolio with derivatives solves in 0.73 s, where the corresponding grid would need $\sim 51^{150} \approx 10^{255}$ states. Solve time grows roughly linearly in N and polynomially in S; S, not N, is the cost driver—the property that lets the engine scale to realistic mandates. Figure 1 plots both.

Table 1: Scenario CVaR-LP solve time vs number of instruments ($S = 4{,}000$).

N (instruments)	LP solve time (s)	Grid states at $m = 51$
5	0.12	$\sim 10^{9}$
10	0.29	$\sim 10^{17}$
20	0.26	$\sim 10^{34}$
30	0.32	$\sim 10^{51}$
50	0.53	$\sim 10^{85}$
75	0.48	$\sim 10^{128}$
100	0.62	$\sim 10^{171}$
150	0.73	$\sim 10^{255}$

Table 2: Scenario CVaR-LP solve time vs number of scenarios ($N = 20$).

S (scenarios)	LP solve time (s)
1,000	0.03
2,000	0.07
4,000	0.21
8,000	0.81
16,000	2.80

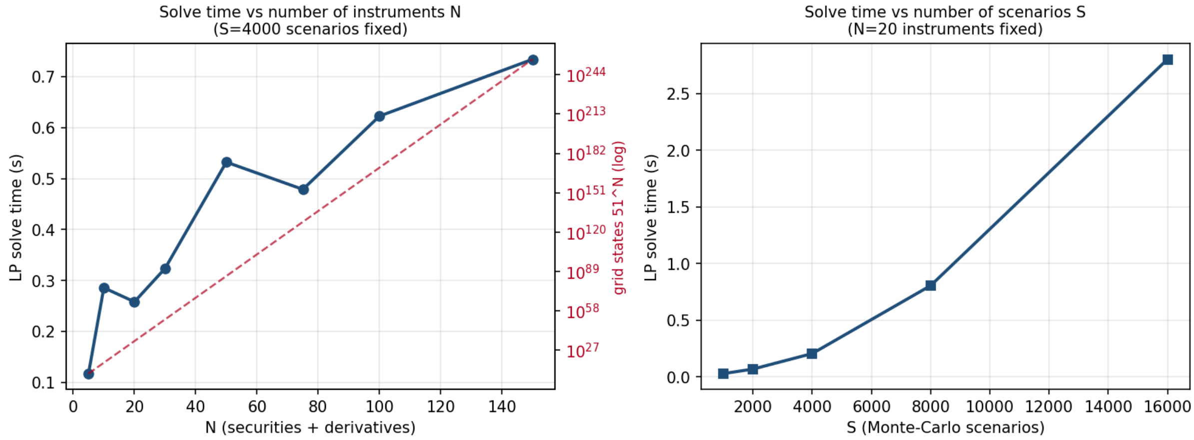

Figure 1: Left: solve time vs N at fixed S, with the 51^N grid state count on a log axis for contrast. Right: solve time vs S at fixed N. Single-thread, scipy-HiGHS, commodity hardware.

9 Benchmarking against standard optimisers

Table 3 is the sanity case: with Gaussian (elliptical) returns and no derivatives, the CVaR-LP optimum lands *exactly* on the mean–variance optimum—identical expected return, volatility and ES—the expected elliptical-distribution equivalence and an independent validation. Equal-weight is dominated; risk-parity trades return for lower dispersion, as designed. Table 4 is the case that motivates the engine: with non-elliptical returns (a Student-$t(3)$ copula, with a protective put and a call available), at a *matched tail-risk budget* the CVaR-LP portfolio reaches materially more expected return than the variance-minimising portfolio held to the same realised ES, because mean–variance is blind to the asymmetry. The magnitude depends on tail-heaviness and the permitted derivative exposure; the robust claim is the direction—CVaR-LP weakly dominates mean–variance at matched tail budget, strictly when returns are non-elliptical.

Table 3: Gaussian, no-derivative ($N = 10$): CVaR-LP coincides with mean–variance.

Portfolio	$\mathbb{E}[r]$	vol	$\mathrm{ES}_{5\%}$	skew
CVaR-LP (floor $L = -20\%$)	11.25%	13.97%	−17.65%	0.03
Mean–variance (matched $\mathbb{E}[r]$)	11.25%	13.97%	−17.65%	0.03
Equal-weight ($1/N$)	8.11%	13.86%	−20.45%	0.00
Risk-parity (ERC)	8.54%	12.02%	−16.21%	0.00

Table 4: Frontier dominance, $t(3)$ copula with put and call available: max $\mathbb{E}[r]$ at a matched realised-ES budget. The mean–variance column is the minimum-variance portfolio held to the same realised-ES floor; it is flat because variance minimisation cannot exploit the asymmetric payoff, whereas the CVaR-LP re-optimises against the tail at each floor.

Tail floor (realised ES)	Mean–variance $\mathbb{E}[r]$	CVaR-LP $\mathbb{E}[r]$	CVaR-LP edge
−30%	7.23%	24.88%	+17.7 pp
−25%	7.23%	20.35%	+13.1 pp
−20%	7.23%	15.82%	+8.6 pp
−17%	7.23%	13.10%	+5.9 pp

10 Robustness and sensitivity

Three sensitivities, run on the scalable engine (five-asset universe, $\alpha = 5\%$). *Copula choice* (Table 5): with identical Normal marginals, the equal-weight $\mathrm{ES}_{5\%}$ deepens monotonically as the copula tail fattens, −24.01% (Gaussian) to −25.65% ($t(3)$)—isolating dependence as a first-order driver of tail risk. *Sample size* (Table 6): the ES estimate is unbiased and its dispersion falls almost exactly as $S^{-1/2}$ (std dev 0.0121 at $S = 1{,}000$ to 0.0022 at $S = 32{,}000$, a 5.5× reduction vs the $\sqrt{32} \approx 5.66$ predicted), confirming the Theorem 1 rate and justifying $S \approx 10^4$ as a default. *Estimation error*: perturbing the mean vector μ by i.i.d. Normal noise and re-optimising, the optimum moves materially—mean weight change (L1) of 0.33, 0.53, 0.73 at input noise 0.5%, 1.0%, 2.0% respectively (out of a 2.0 maximum), with optimal-return spread 0.51%, 0.92%, 1.72%. This is the Markowitz "error-maximisation" effect, inherited because the objective is linear in μ; it is the numerical case for shrinkage or robust estimators in deployment. In short: tail risk is driven first-order by the copula and by μ estimation error, and only second-order by Monte-Carlo noise once $S \gtrsim 10^4$.

Table 5: Copula sensitivity (equal-weight $\mathrm{ES}_{5\%}$).

Copula	$\mathrm{ES}_{5\%}$
Gaussian	−24.01%
Student-t ($\nu = 8$)	−24.45%
Student-t ($\nu = 5$)	−24.84%
Student-t ($\nu = 3$)	−25.65%

Table 6: Monte-Carlo error vs S (20 seeds).

S	ES mean	ES std dev
1,000	−24.27%	0.0121
2,000	−24.24%	0.0062
4,000	−24.28%	0.0054
8,000	−24.30%	0.0042
16,000	−24.30%	0.0028
32,000	−24.29%	0.0022

11 Results: derivative overlays

Table 7 reports a single overlay under the grid (VaR) engine on the three-asset thesis data set (annual returns 5/10/25%, volatilities 5/20/50%, with the mid- and high-risk assets correlated at 0.4 and the low-risk asset uncorrelated). An at-the-money straddle lifts the optimal expected return from 9.76% to 11.29% at the same −10%/5% limit, lifting skewness from 0 to +0.99; the gain is paid in variance and collected in positive skewness—the dimension mean–variance cannot see. Table 8 reports the scalable (α-CVaR) engine: a protective put (struck 0.8×, one year) lifts the optimum from 11.00% to 15.73% at a −20% CVaR floor, raising skewness to +0.71. By Proposition 2 the two tables answer different questions and are kept separate.

Table 7: Grid engine, VaR constraint ($H = -10\%$, $\alpha = 5\%$), three-asset thesis data.

Overlay	$\mathbb{E}[r]$	$\Delta\mathbb{E}[r]$	skew
none	9.76%	—	0
ATM straddle	11.29%	+1.53 pp	+0.99

Table 8: Scalable engine, α-CVaR floor ($L = -20\%$), same data.

Overlay	$\mathbb{E}[r]$	$\Delta\mathbb{E}[r]$	skew
none	11.00%	—	≈ 0
protective put (0.8×, 1y)	15.73%	+4.73 pp	+0.71

12 Out-of-sample backtesting across regimes

The backtest builds optimal weights on a construction window, then holds those fixed weights (buy-and-hold) through a later evaluation window, marking any derivative to market with a shrinking maturity, and compares expected with realised behaviour against a benchmark. Earlier versions of this work reported a single 2017 window; we now report three regimes (Table 9) on AAPL/MSFT/JPM versus the S&P 500, with the construction window immediately preceding each evaluation year. The central, defensible finding is that *the downside constraint ($\mathbb{P}(r < -10\%) \leq 5\%$) held out-of-sample in every regime, including the 2020 COVID crash:* no window finished below the -10% threshold. Realised returns are highly regime-dependent (15% to 46%) and the expected-vs-realised gap flips sign—realised above expectation in 2017 and 2020, below in 2018—so the 2017 result is correctly read as one draw, not evidence of systematic outperformance. Realised alpha against the S&P is positive in all three windows, but with only three windows and varying fit this is illustrative, not statistically significant. We therefore read this section as a *demonstration* that the constructed portfolios behave as designed out of sample—the downside floor holds and the overlay buys asymmetry—rather than as inferential evidence of outperformance; establishing the latter would require many more windows, broader universes and a formal test, left to future work.

Table 9: Out-of-sample backtest, no-derivative portfolio (P1), AAPL/MSFT/JPM vs S&P 500, $r_f = 3\%$, VaR $H = -10\%/\alpha = 5\%$. Live data via the companion app.

Eval	Regime	Expected	Realised	Real. vol	vs H=−10%	α vs S&P	β
2017	strong trend	23.31%	28.19%	14.30%	+28.08% (no breach)	+1.99%	1.47
2018	Q4 selloff	24.44%	14.95%	27.36%	+14.89% (no breach)	+28.65%	1.42
2020	COVID crash	27.02%	46.35%	43.47%	+46.31% (no breach)	+26.01%	1.12

Derivative overlays in the backtest

Two overlays illustrate the asymmetry of derivative effects under a static hold (Table 10). A *protective put* (strike 0.90) earns near-zero optimal weight in these equity universes under the VaR constraint, so the with-derivative portfolio (P2) tracks the no-derivative portfolio (P1) in 2017 and 2018; in 2020—a year that crashed then fully recovered—a held put cost about 6.9 percentage points of realised return, exactly the premium drag a static hedge incurs when the crash does not dominate the window. By contrast a *geared-upside call*—a deep in-the-money call on the highest-volatility holding (AAPL)—is held in size by the optimiser and amplifies the strong 2017 trend: at strike 0.70 the optimal weight is 13.9% and P2 realises 46.67% versus P1's 28.19%

(+18.5 pp), with realised α of +11.57% and β rising to 1.76, while still respecting the −10% downside floor. This is the configuration behind the large derivative-enhanced backtest figures: a geared call on the trending name, not the protective put. The magnitude is strike-dependent (strike 0.80 gives weight 7.2%, $\mathbb{E}[r]$ uplift to 41.20% and α +8.27%), and a single window is one draw; we report it as a demonstration of mechanism, not evidence of outperformance.

Table 10: Derivative overlays, 2017 window unless noted (AAPL/MSFT/JPM, S&P 500 benchmark).

Overlay	Optimal wt	P1 realised	P2 realised	P2 α vs S&P
Protective put (0.90), 2017	$\approx$ 0%	28.19%	28.13%	+1.94%
Protective put (0.90), 2018	$\approx$ 0%	14.95%	15.36%	+28.85%
Protective put (0.90), 2020	small	46.35%	39.43%	+20.95%
ITM call on AAPL (0.80), 2017	7.2%	28.19%	41.20%	+8.27%
ITM call on AAPL (0.70), 2017	13.9%	28.19%	46.67%	+11.57%

Static option holding. The optimiser is single-period: it chooses fixed weights for a one-year horizon and the backtest holds them, marking each derivative to market with a shrinking maturity but never re-optimising or re-hedging within the window. For linear holdings this is mild, but for options it is a genuine limitation. An option's delta, gamma and theta change materially as spot moves and maturity decays, so an inception-optimal position is generally not optimal mid-horizon; a static hold leaves gamma unmanaged and pays full time decay. The protective put's near-zero weight and the call's strike-sensitivity in Table 10 are both symptoms of this: a buy-and-hold option is a blunt instrument. The single-period optimum should be read as the inception allocation of an annually-rebalanced mandate; multi-period re-optimisation and delta-hedging are the natural next step.

Transaction costs

Because the strategy is buy-and-hold—one trade at inception, with the derivative marked to market rather than re-traded—transaction costs enter only once, at construction, and are therefore second-order for the figures above. As an illustration, a one-time round-trip cost of 10 basis points on the initial notional lowers each window's realised annual return by about 0.1 percentage point, leaving every qualitative conclusion (downside respected, regime-dependent returns, the call's uplift) unchanged. The materially cost-sensitive setting is a *rebalancing* mandate, where turnover recurs each period; modelling per-trade costs and a turnover penalty (an additional linear term, already supported by the LP in (4)) is part of the multi-period extension and is left to future work. We flag this explicitly so the buy-and-hold results are not mistaken for a frictionless rebalancing claim.

13 Changes and corrections relative to the 2012 thesis

(a) Expected-Shortfall enforcement. In the original R code the gradient-refinement step maximises return under a VaR penalty, and the ES condition enters only at grid-eligibility—so the reported "ES-constrained" optima are not strictly ES-feasible after refinement. The app reproduces this thesis-faithful behaviour by default for comparability, and adds a rigorous-ES mode that maximises return strictly subject to the ES floor and stays ES-feasible throughout. Validated against an independent brute-force ES maximum, a put overlay at $L = -15\%$ attains 15.5% under the rigorous mode versus 13.2% under the thesis-faithful method (+2.3 pp), and a straddle overlay at $L = -20\%$ attains 20.3% versus 11.9% (+8.4 pp); the thesis-faithful refinement can finish ES-infeasible.

(b) Fair pricing of structured notes. An earlier headline for a capital-guaranteed note overstated its benefit because the note was valued at par. Pricing every instrument with Black–Scholes (a guaranteed note becomes the present value of its bond floor plus its embedded option) shrinks a par-priced headline of roughly +23% to about +0.7 to +0.8 pp, and shows the fairly-priced note is mean–variance-dominated yet can still improve the downside-constrained objective. That is a genuine behavioural effect rather than a pricing artefact.

14 Results provenance

Table 11 tags every headline result as replication, correction or extension, and names the risk floor each uses—so a reader can see at a glance what is reproduced, what is corrected, and what is new, and the two tail measures are never conflated.

Table 11: Provenance of headline results.

Result	Type	Engine / method	Risk floor
3-asset base case 9.78%	Replication	Grid, VaR	$\mathbb{P}(r < H) \leq \alpha$
ATM straddle +1.53 pp	Extension	Grid, VaR	$\mathbb{P}(r < H) \leq \alpha$
Rigorous-ES +2.3 to +8.4 pp	Correction	Grid, rigorous ES	$\mathbb{E}[r \mid r < H] \geq L$
Fair-priced note +0.7 to +0.8 pp	Correction	Grid, fair pricing	—
Protective put +4.73 pp	Extension	Scenario CVaR-LP	α-CVaR $\geq L$
Scaling to $N = 150$ in 0.73 s	Extension	Scenario CVaR-LP	α-CVaR $\geq L$
Downside held in 2017/18/20	Extension	Backtest (live)	$\mathbb{P}(r < H) \leq \alpha$
ITM call overlay 2017 +18.5 pp	Extension	Backtest (live)	$\mathbb{P}(r < H) \leq \alpha$

15 Relation to existing methods

The scenario-LP machinery is the Rockafellar–Uryasev (2000, 2002) CVaR linear program, applied to portfolio choice by Krokhmal, Palmquist & Uryasev (2002); the Gaussian, derivative-free reduction to mean–variance is the elliptical-distribution equivalence. What is new is the composition: a goal-based, mental-account (Das–Statman) objective, solved as a CVaR LP, with arbitrage-consistent per-scenario derivative pricing so that asymmetric overlays enter the same convex program. Concretely, standard CVaR portfolio optimisation does not carry a goal-based mental-account structure or price derivatives consistently inside each scenario, and existing mental-account models are solved on exhaustive grids that do not scale past a few assets and a single derivative; the engine here is, to our knowledge, the first to combine all three in one program that scales to realistic universes. Relative to robust optimisation (Goldfarb & Iyengar, 2003; Ben-Tal, El Ghaoui & Nemirovski, 2009) and distributionally robust optimisation (Mohajerin Esfahani & Kuhn, 2018), which immunise against parameter/distribution ambiguity, our engine takes the distribution as given and solves the behavioural tail-constrained problem at scale; the two are complementary, and a Wasserstein-DRO wrapper around the same scenario set is a natural extension. Against the naive $1/N$ benchmark (DeMiguel, Garlappi & Uppal, 2009) the engine is, like all optimisers, exposed to estimation error (Section 10), which is why shrinkage (Ledoit & Wolf, 2004) belongs in any deployment.

16 Limitations

The grid engine assumes multivariate-normal marginals and is exact only on small universes ($n \leq 4$); the scalable engine is approximate, with Monte-Carlo error of order $S^{-1/2}$ that is heavier in the tail and is mitigated by sample size, antithetic variates and common random numbers. Results depend on scenario quality: the copula choice and the estimation of μ, σ, Σ drive the answer, and estimation error worsens with n, so shrinkage or robust estimators belong in any production pipeline. Derivatives are priced under flat-volatility Black–Scholes, ignoring the volatility smile, skew and jumps that drive real structured-product value; because each scenario is priced from its own simulated underlying, the framework can host a local-volatility, smile-aware or jump pricer at the cost of one function call per scenario, with the LP unchanged. The model is single-period; the backtest bridges to a real holding window but does not re-optimise within it. The scalable engine exposes a subset of the instrument library. None of these is a barrier to the construction; each is a calibration responsibility.

17 Implementation

The optimiser is written in Python (NumPy, SciPy, the HiGHS linear-program solver and copula sampling) behind a Streamlit interface. The numerical pipeline is open and reproducible: copula scenarios → Black–Scholes pricing → CVaR linear program → efficient frontier → interactive app and PDF report. The scaling, benchmark and robustness figures in Sections 8–10 were produced by the prototype engine; the backtest of Section 12 by the live application on Yahoo Finance data. The code and this paper are available at `https://github.com/SamiJeddou/behavioral-portfolio-optimizer`, and the interactive application is live at `https://sami-jeddou-behavioral-portfolio-optimizer.streamlit.app`.

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